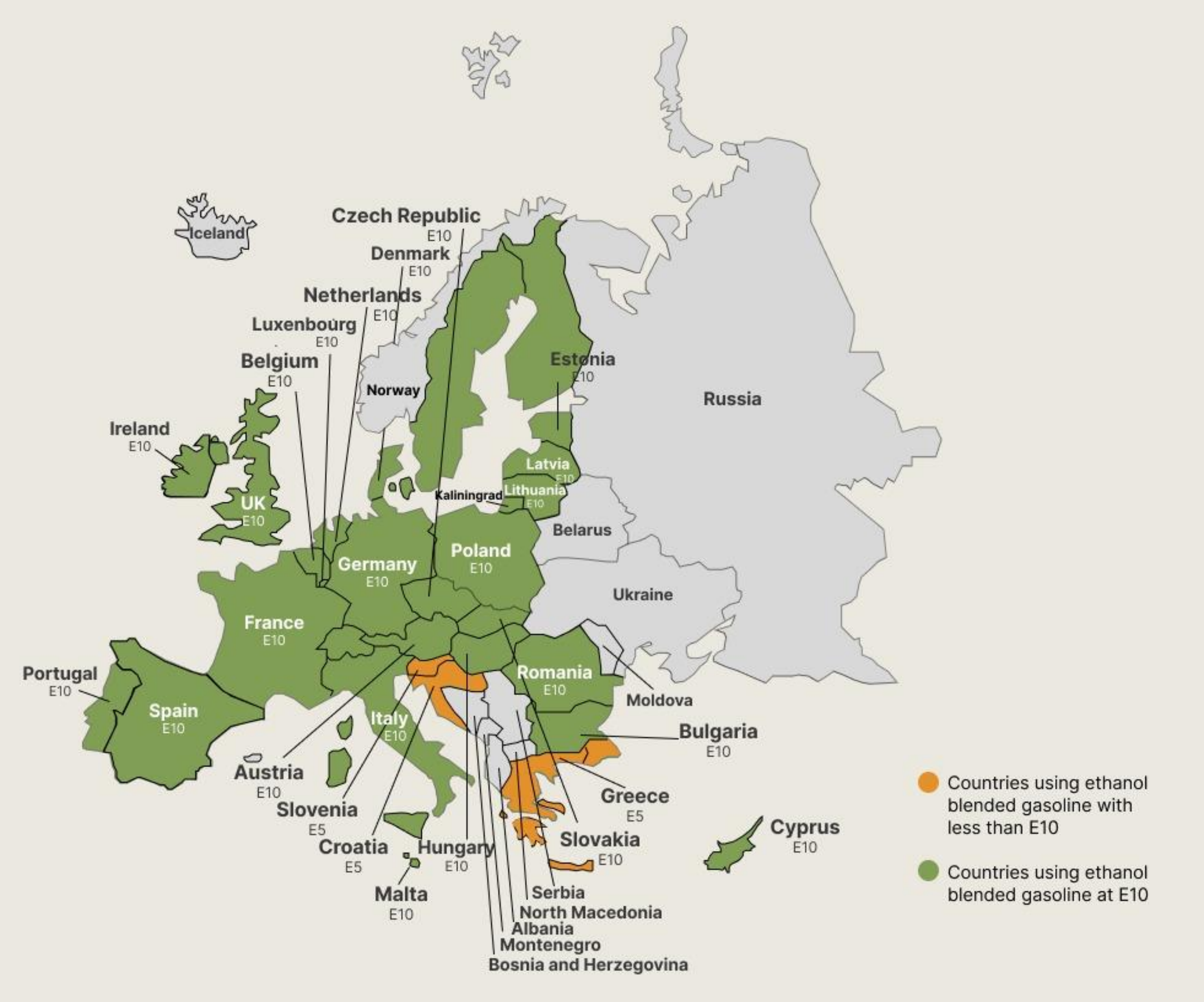
Ethanol Blending in Gasoline - Slovenia

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

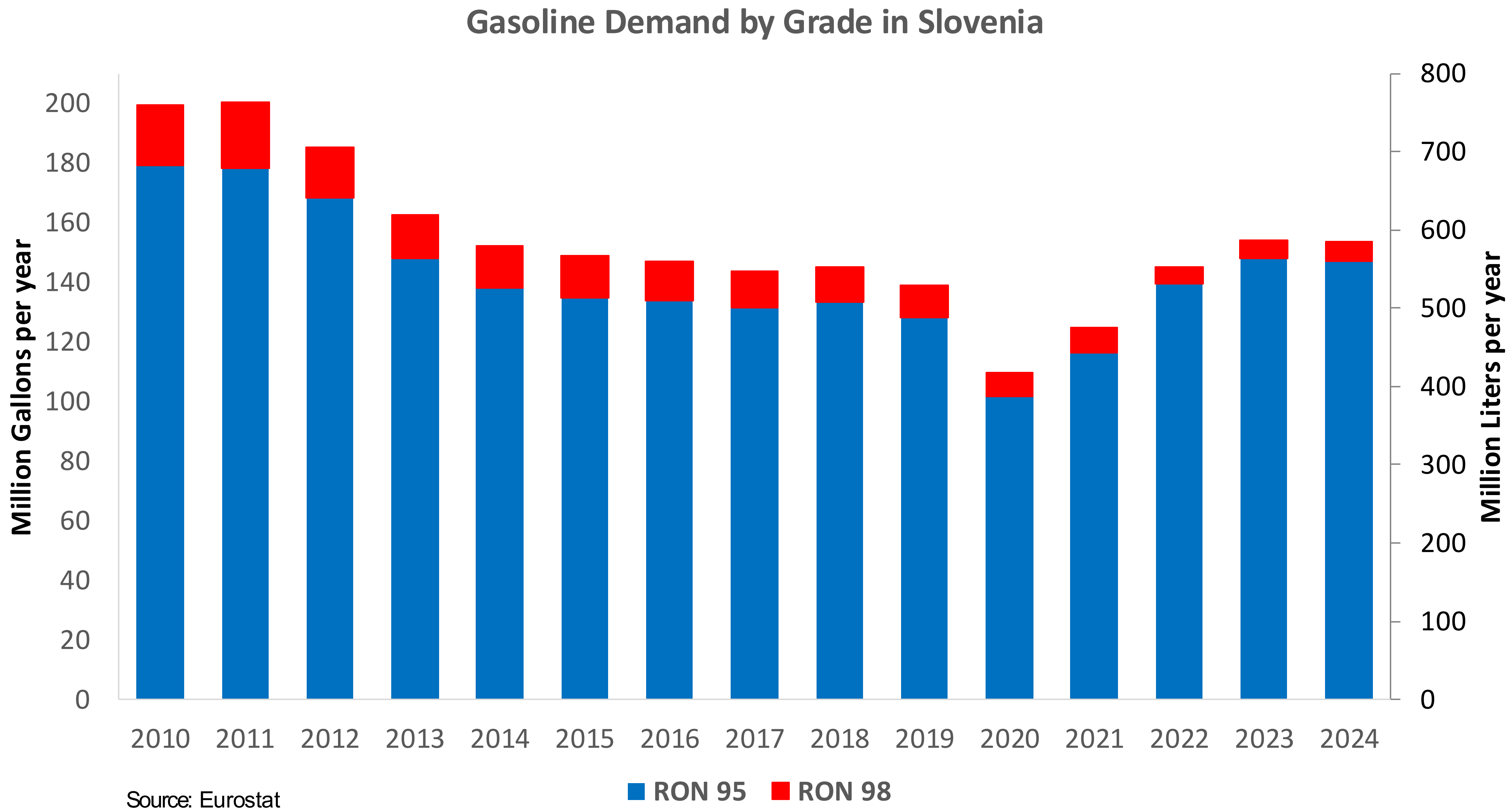
Ethanol Gasoline Blending in Slovenia

In 2024, gasoline consumption in Slovenia was 585 million liters (155 million gallons) and growth rate was 2.0%. Gasoline production and net imports were and 574 million liters (and 152 million gallons) correspondingly. Slovenia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.1 RON.

In 2024, ethanol consumption in Slovenia was 3 million liters (1 million gallons) representing .5% of gasoline demand. Ethanol production and Net Imports were and 3 million liters (1 and 1 million gallons) correspondingly. Ethanol source is Cereals 100%.

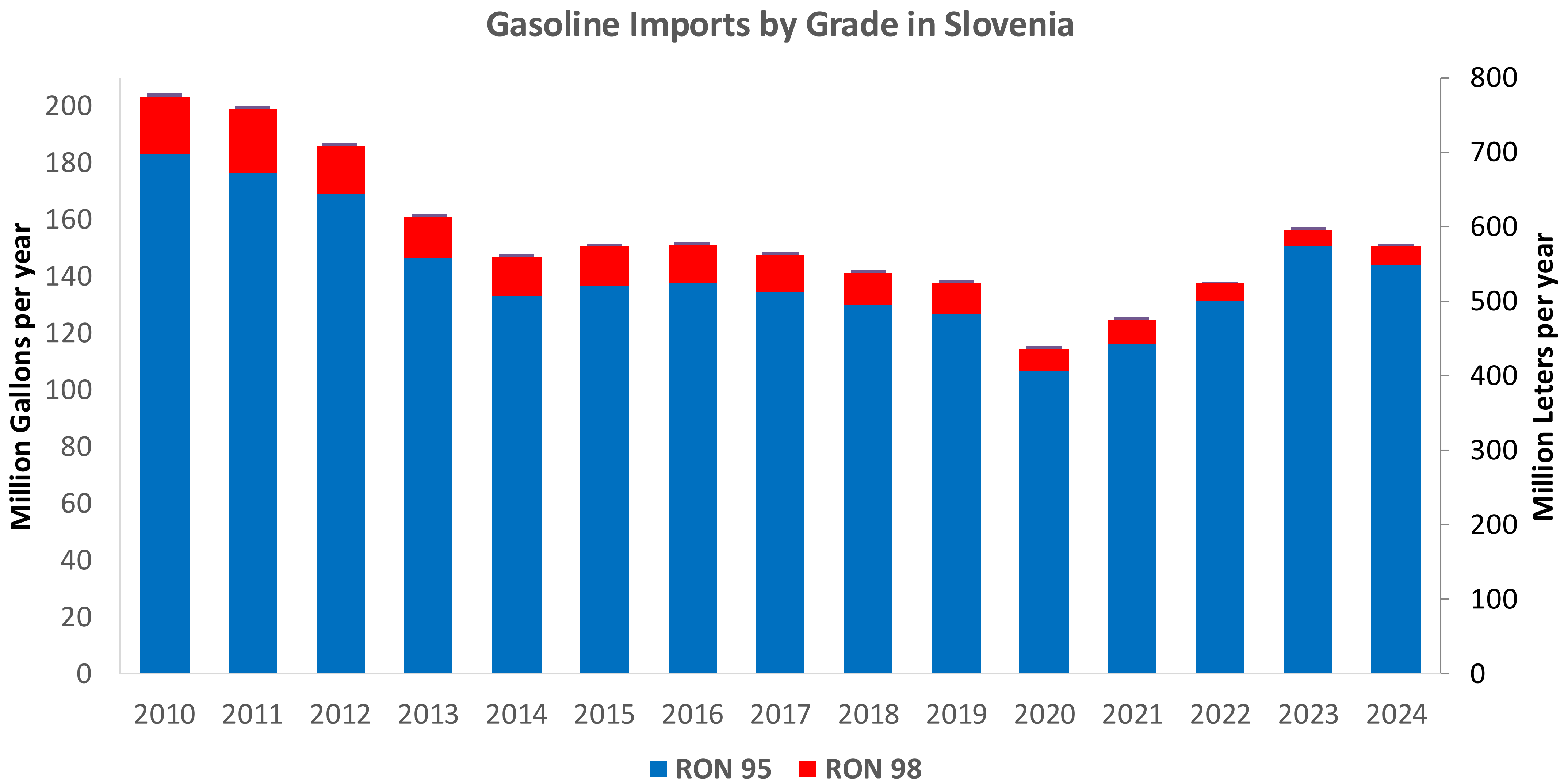
Source: Eurostat, European Commision

Gasoline Demand in Slovenia



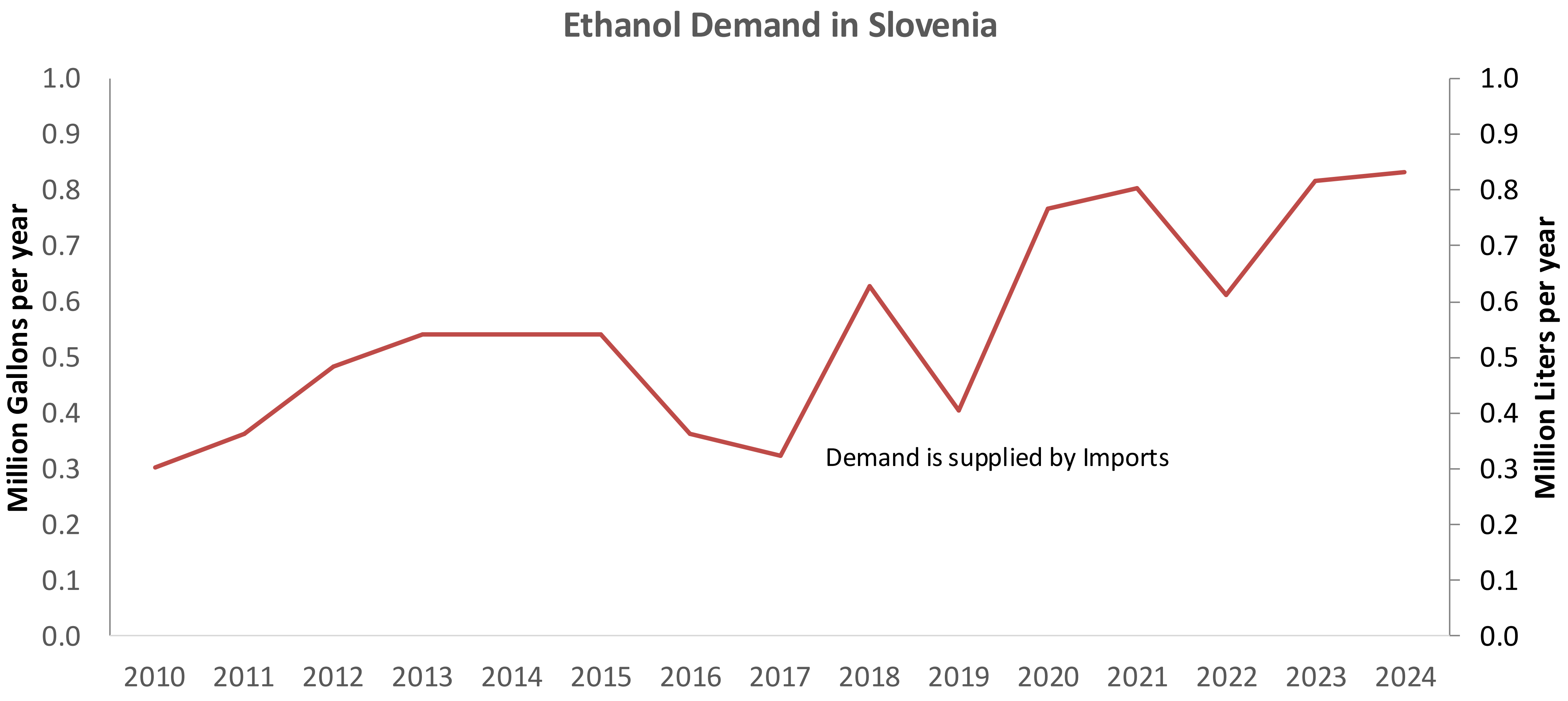
Source: Eurostat

Gasoline Imports in Slovenia



Source: Eurostat

Ethanol Balance in Slovenia



Source: Eurostat

Gasoline Quality in Slovenia

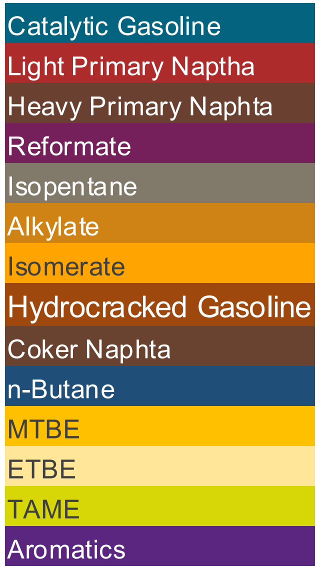
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.2							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

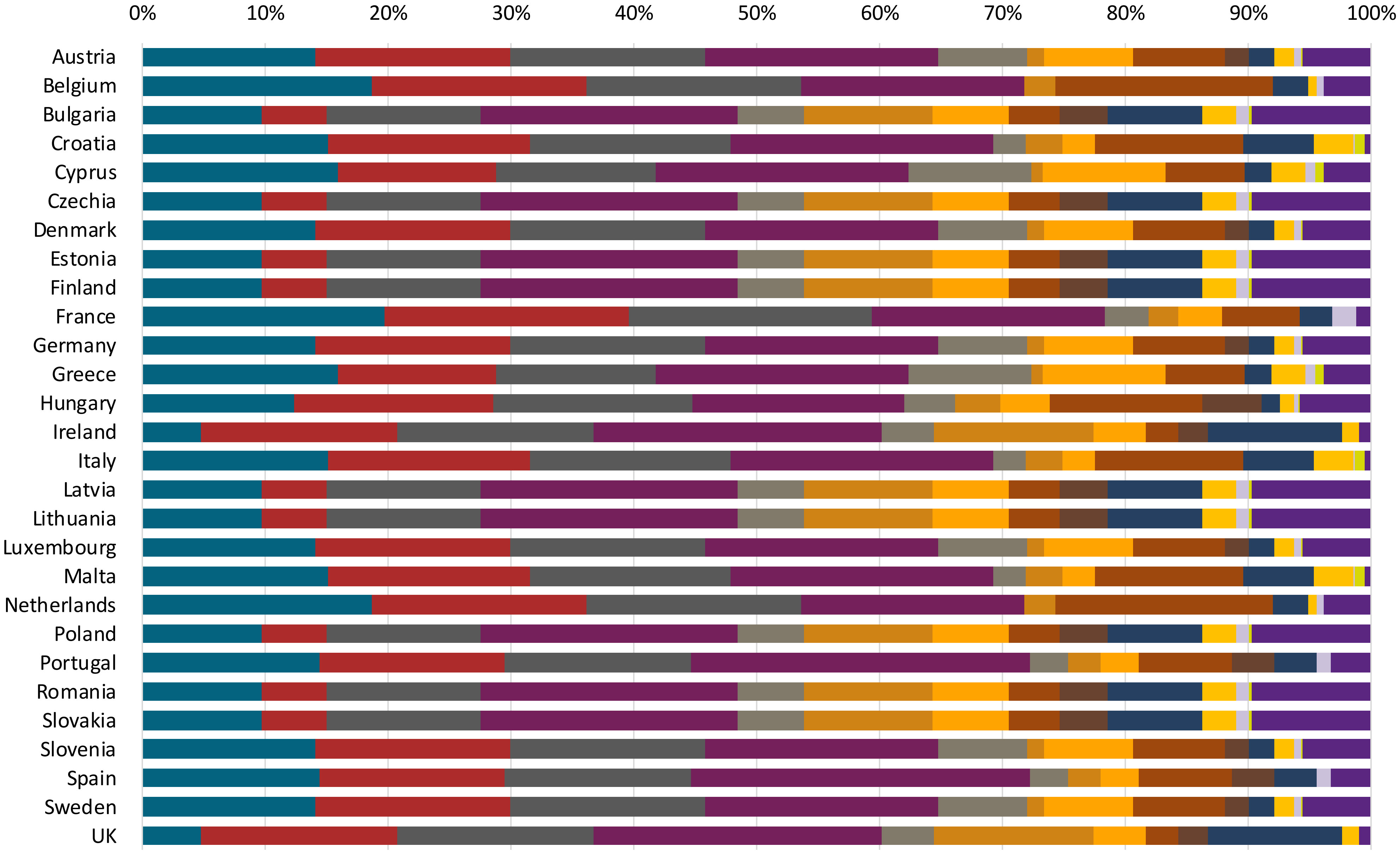
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:





Source: Faro90

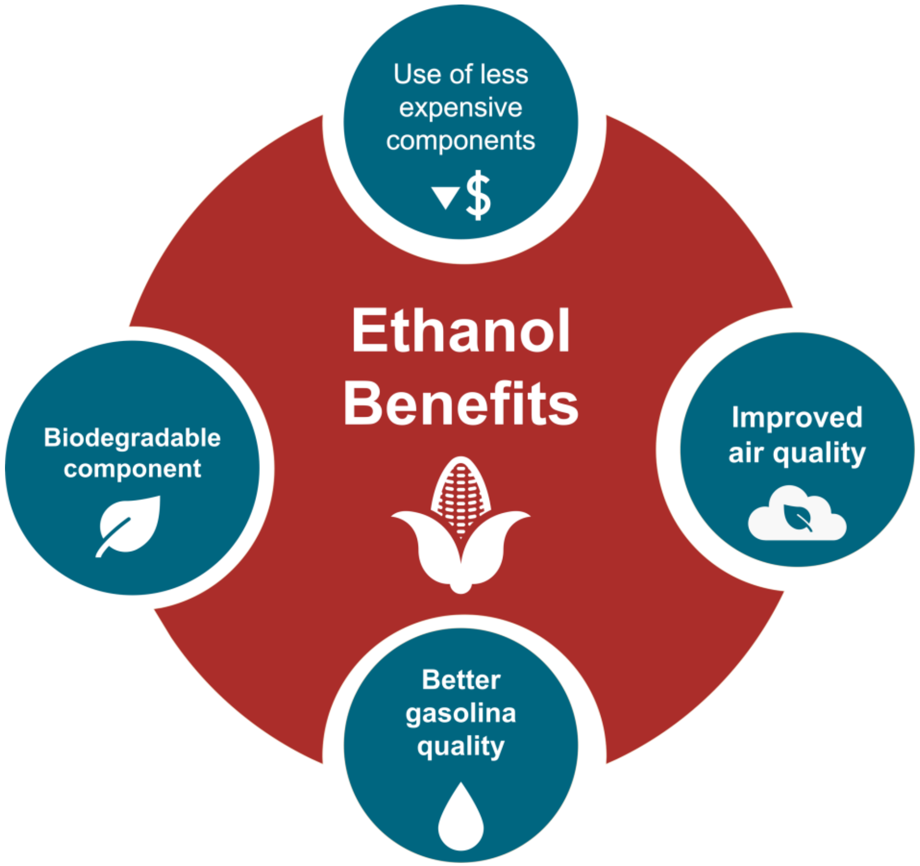
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

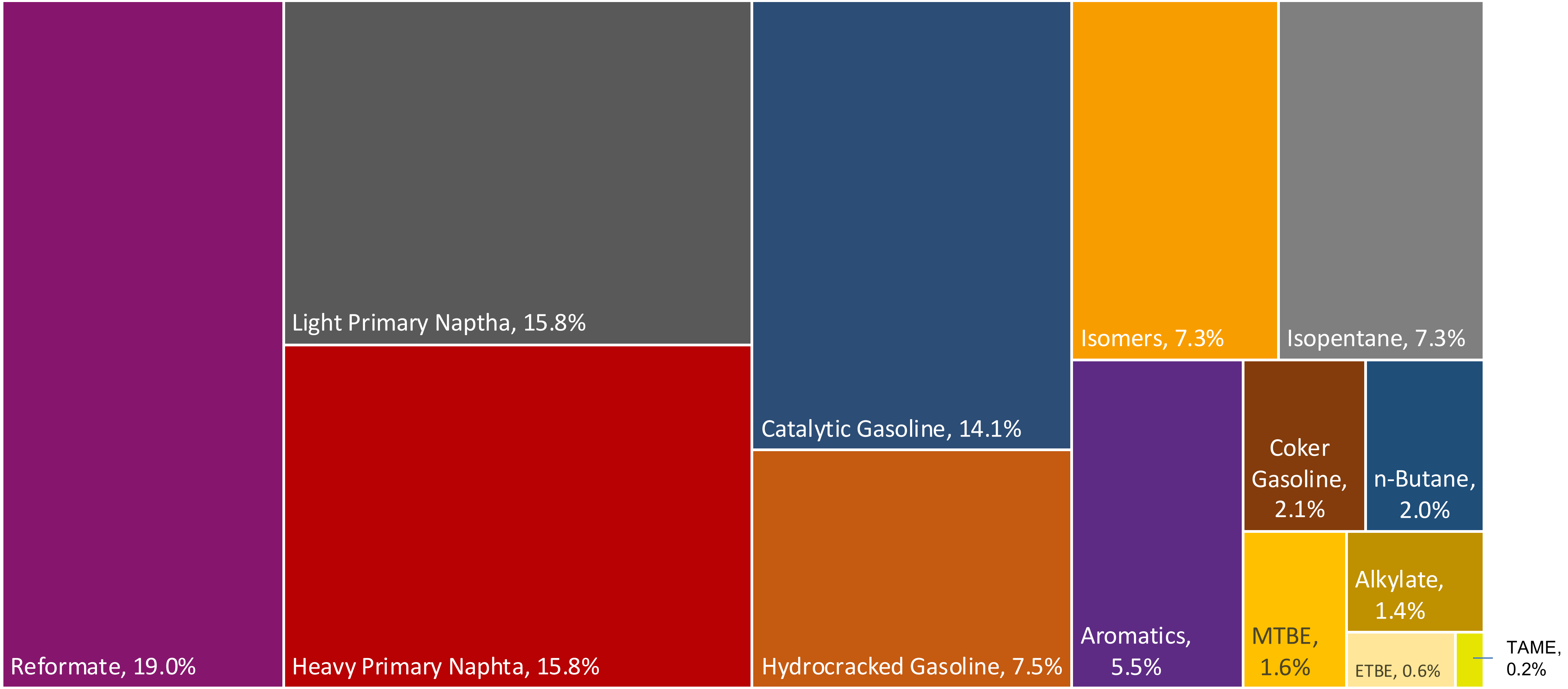
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

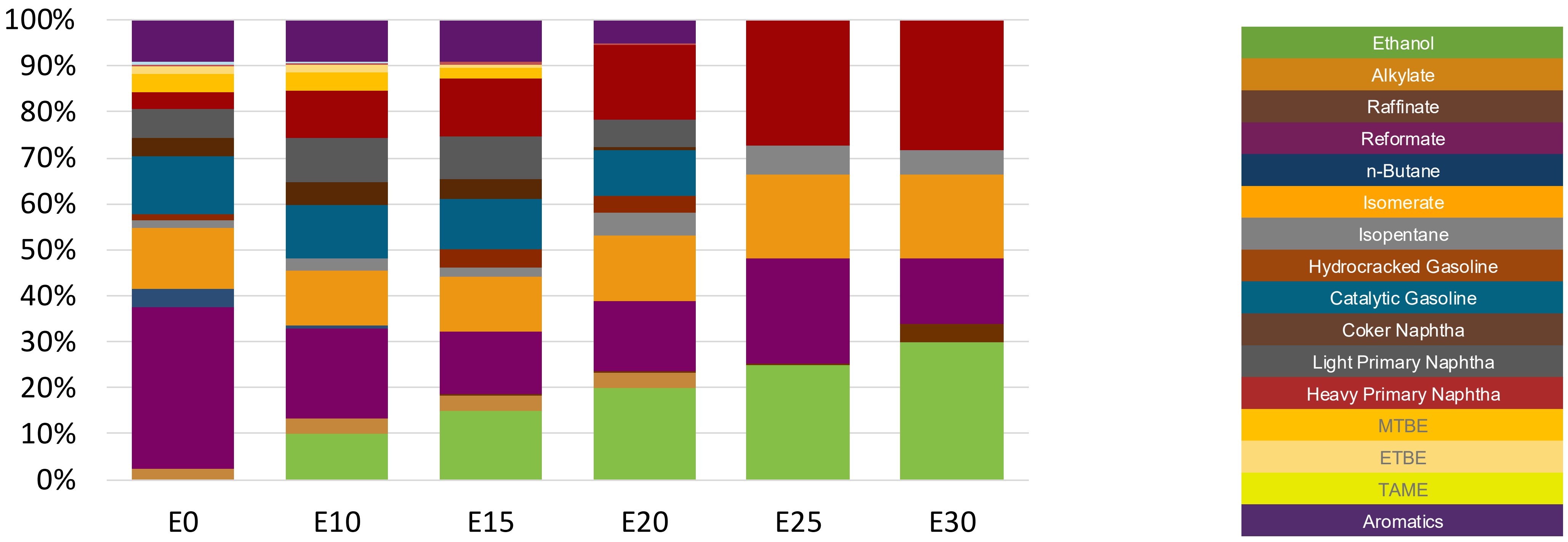


Slovenia Available Blending Components



Source: Faro90

Slovenia – Gasoline 95 – Constant Octane

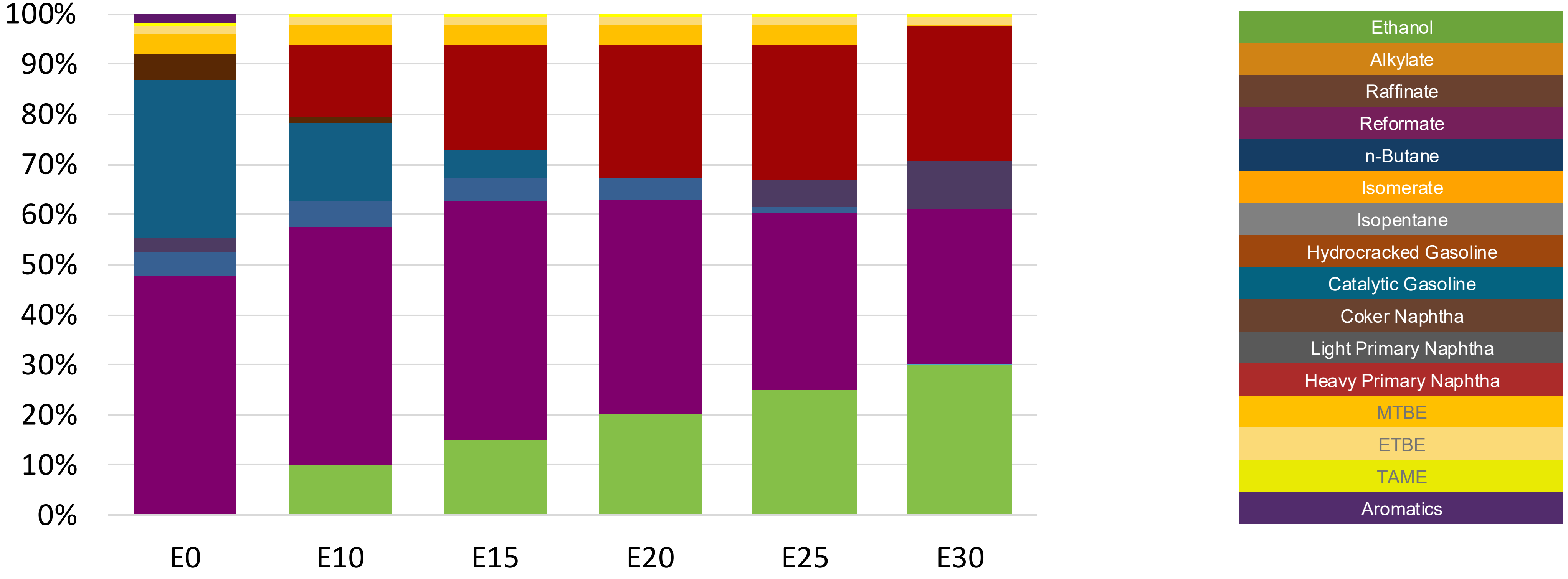


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.292	$ 2.164	$ 2.103	$ 2.022	$ 1.900	$ 1.834

Average CIF 2024 Prices

Source: Faro90

Slovenia - Gasoline 98 - Constant Octane

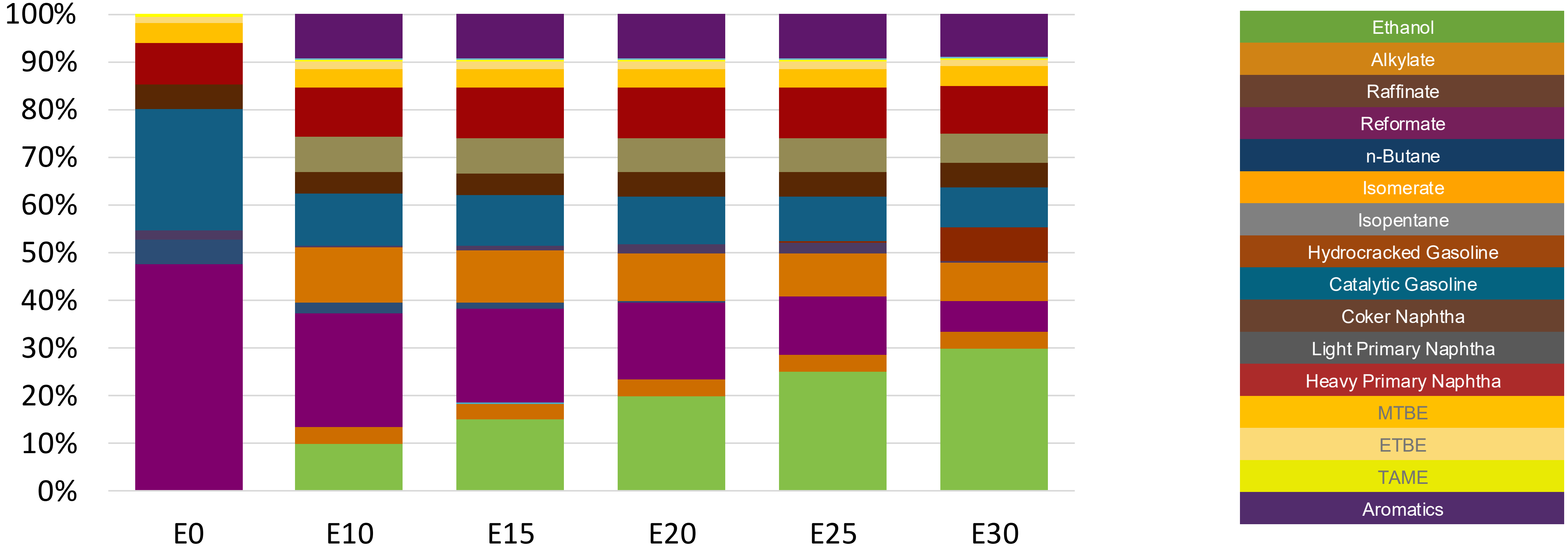


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.367	$ 2.238	$ 2.166	$ 2.089	$ 2.012	$ 1.945

Average CIF 2024 Prices

Source: Faro90

Slovenia – Gasoline 95 – Increased Octane

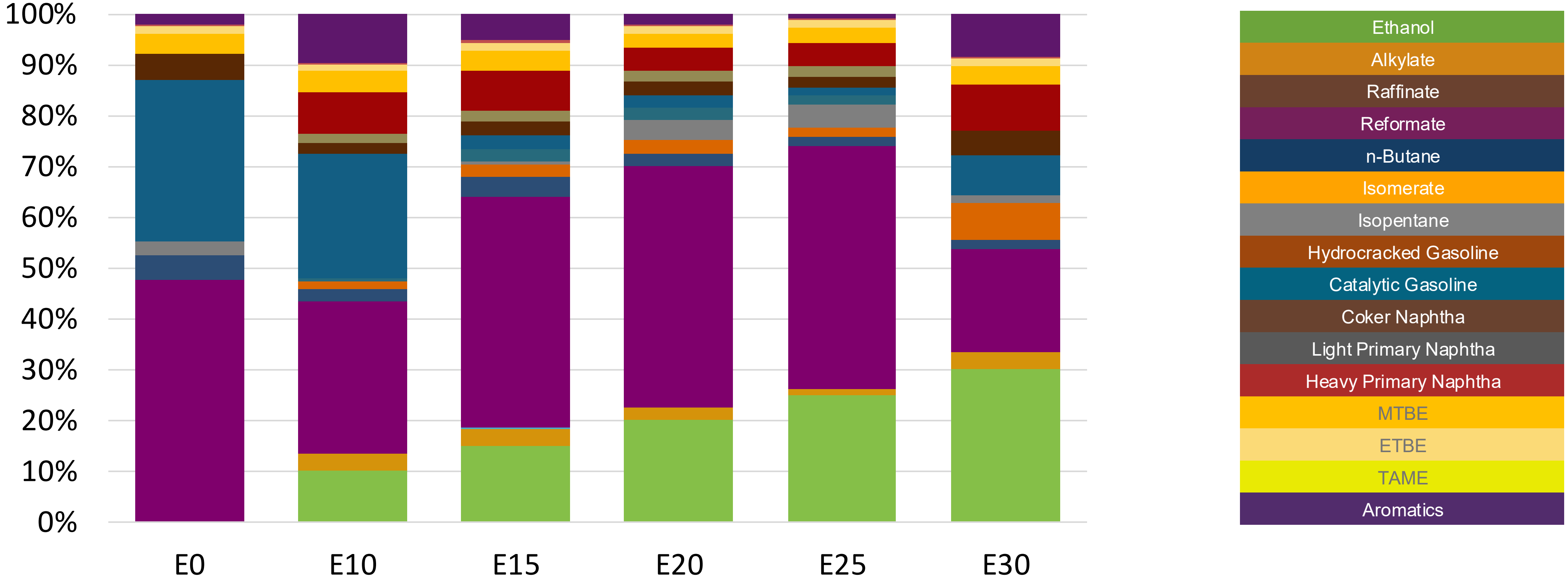


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.229	$ 2.229	$ 2.229	$ 2.229	$ 2.229	$ 2.229

Average CIF 2024 Prices

Source: Faro90

Slovenia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.367	$ 2.367	$ 2.367	$ 2.367	$ 2.367	$ 2.367

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

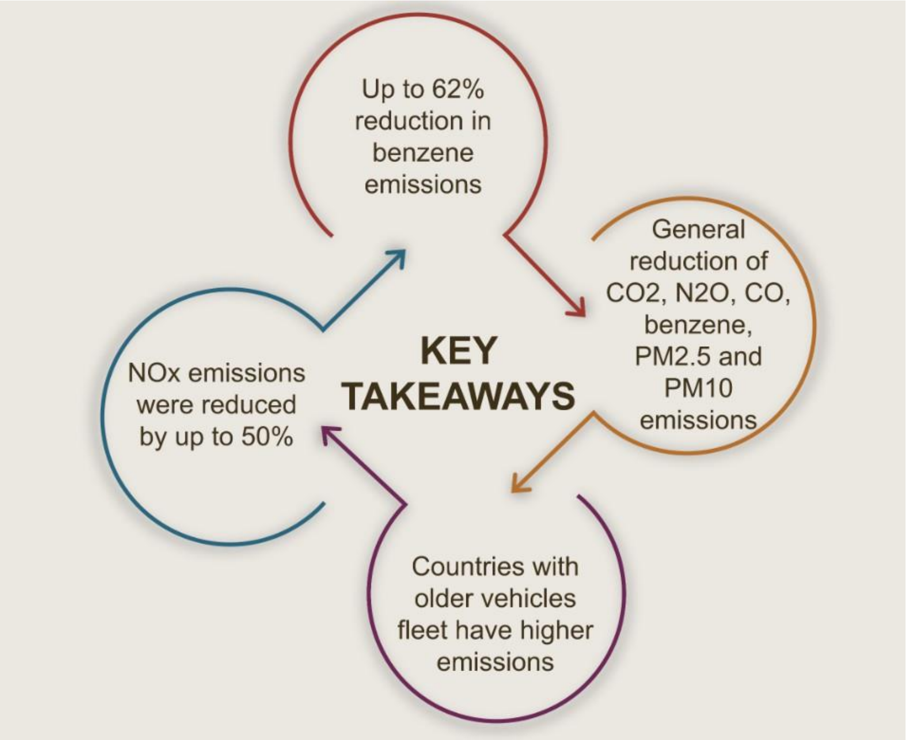
The model used in this analysis takes as a reference the **[International Vehicle Emissions Model (IVE).]**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - Slovenia

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	97
		Single FI	3-Ways		>8 years	6,881
		Multi FI	3-Ways		>8 years	166,459
			Euro III		>8 years	161,409
			Euro IV		>8 years	316,395
			Euro V		>8 years	90,173
			Euro 6		>8 years	208,911
					4-8 years	271,329
					<4 years	1,036,869
			Hybrid		All	7,870
		Electric			All	11,534
	Medium	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	1,394
		Multi FI	3-Ways		>8 years	6,675
			Euro III		>8 years	11,944
			Euro IV		>8 years	26,760
			Euro V		>8 years	9,211
			Euro 6		>8 years	2,717
					4-8 years	15,915
					<4 years	32,026
			Hybrid		All	28,790
		Electric			All	38
Sml Engines	Light	4-Cycle	None	PCV	>8 years	14,421
			Euro III		>8 years	8,960
			Euro IV		>8 years	23,316
			Euro V		>8 years	6,915
			Euro 6		>8 years	15,676
					4-8 years	36,500
					<4 years	144,108
			Hybrid		All	0
		Electric			All	0
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	174
		Fuel-Injection	3-Ways		>8 years	133,707
			Euro III		>8 years	11,114
			Euro IV		>8 years	35,475
			Euro V		>8 years	10,440
			Euro 6		>8 years	45,655
					4-8 years	50,135
					<4 years	239,718
			Hybrid		All	214
		Electric			All	372

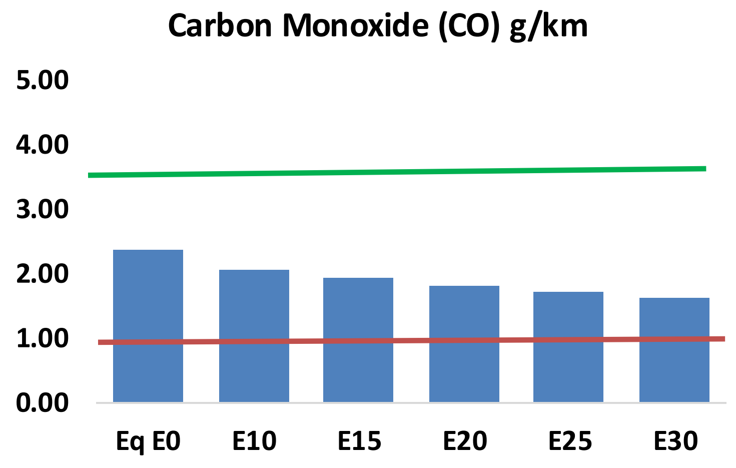
0 200,000 400,000 600,000 800,000 1,000,000 1,200,00

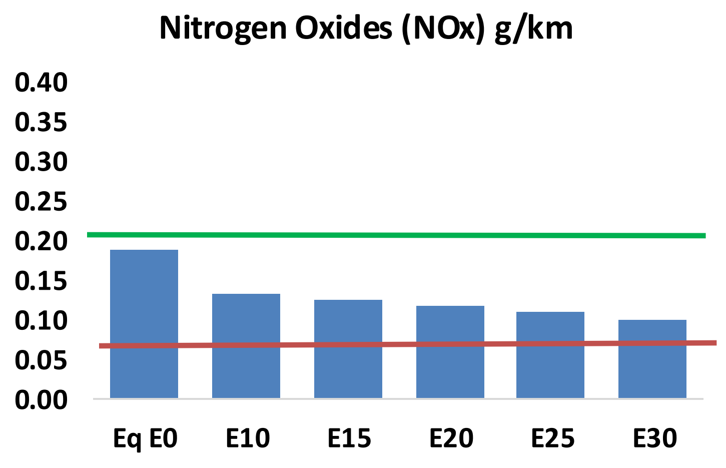
Vehicle Fleet: 3,190,297
Gasoline Vehicle Fleet: 1,163,863

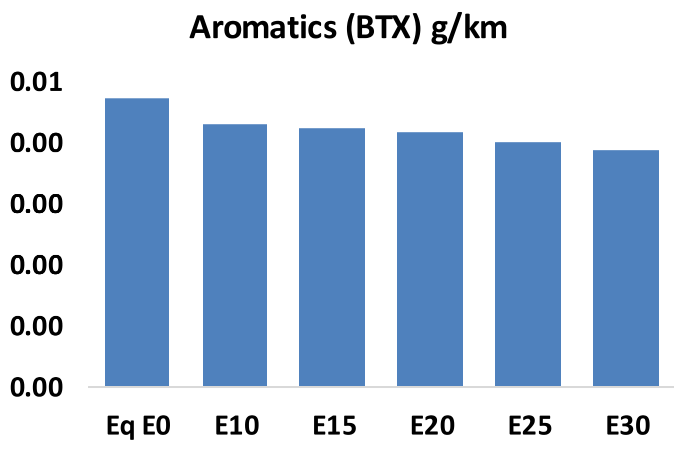
Source: Eurostat, ACEA

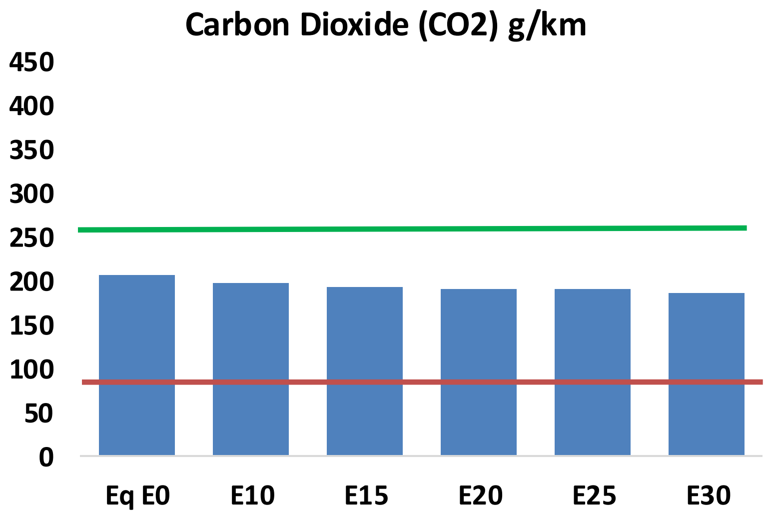
Slovenia – Vehicular Emissions

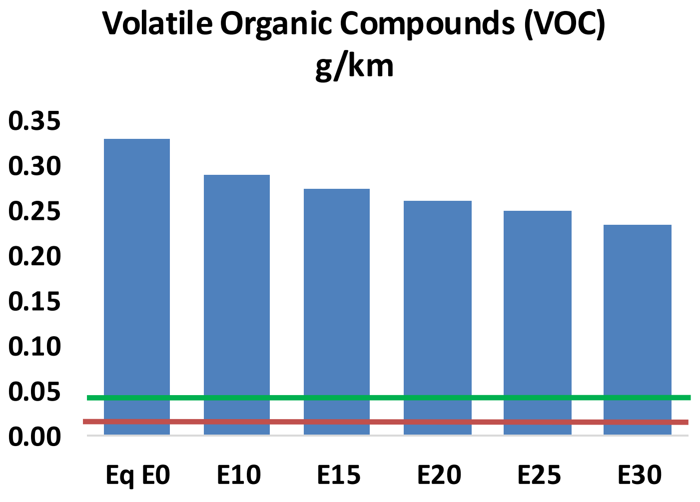
TIER III BIN 125 United States
Euro 6 Standard

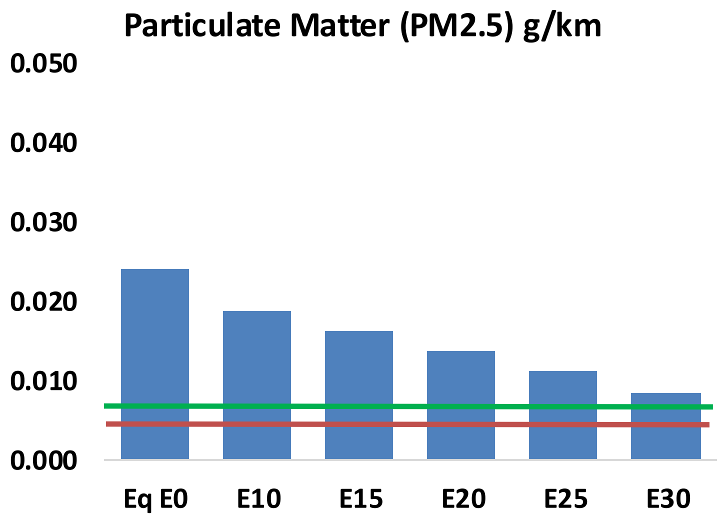












Source: Faro90

Slovenia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	66,460	62,046	57,631	54,148	50,707	48,079	45,138	-13%	-24%
CO2	5,814,866	5,665,340	5,524,122	5,413,058	5,358,100	5,329,401	5,231,165	-5%	-8%
VOC	9,254	8,696	8,137	7,709	7,293	6,976	6,539	-12%	-21%
NOx	5,327	3,996	3,729	3,515	3,315	3,094	2,861	-30%	-38%
SOx	65	60	55	51	47	43	38	-15%	-28%
NH3	2,043	2,023	2,003	2,012	2,035	2,051	2,064	-2%	0%
N2O	61	61	60	61	62	63	63	-1%	2%
CH4	2,030	2,030	2,030	2,061	2,105	2,144	2,180	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	63	61	59	57	57	56	55	-6%	-9%
Acetaldehyde	160	183	269	410	558	638	754	68%	249%
Formaldehyde	545	529	617	725	759	840	914	13%	39%
Benzene	132	127	120	119	118	112	109	-9%	-11%
PM2.5	371	329	288	250	212	172	130	-22%	-43%
PM10	303	269	235	204	173	141	107	-22%	-43%

Source: Faro90

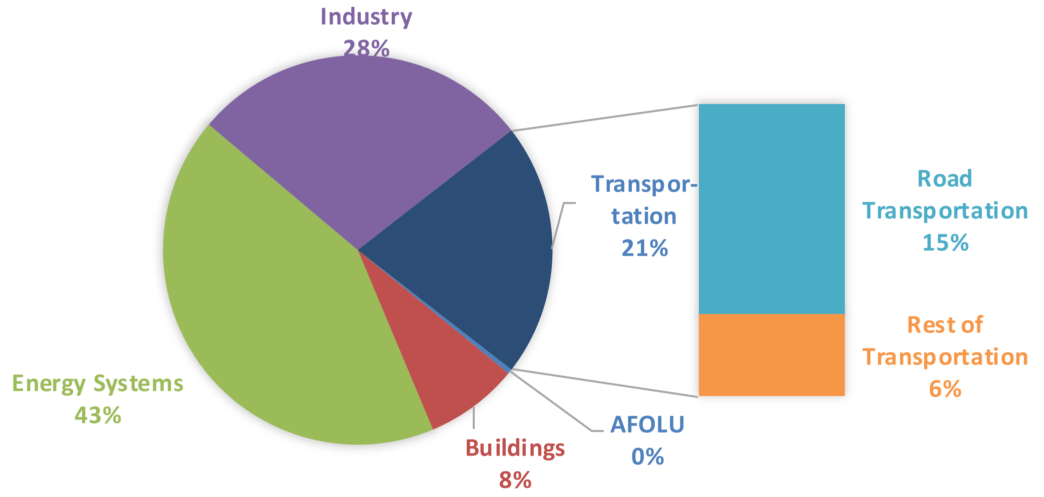
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

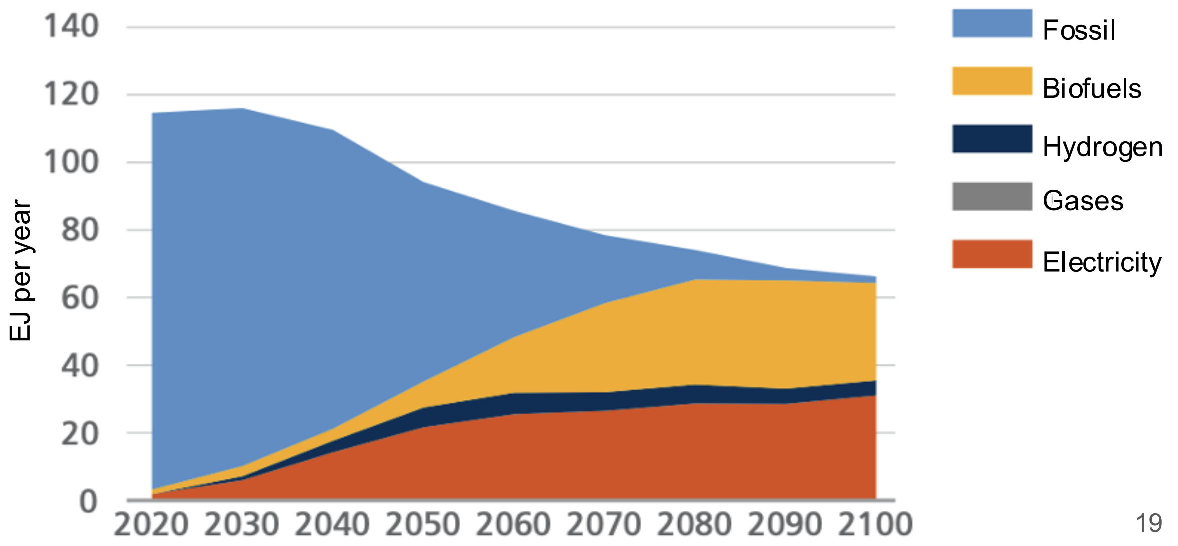
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

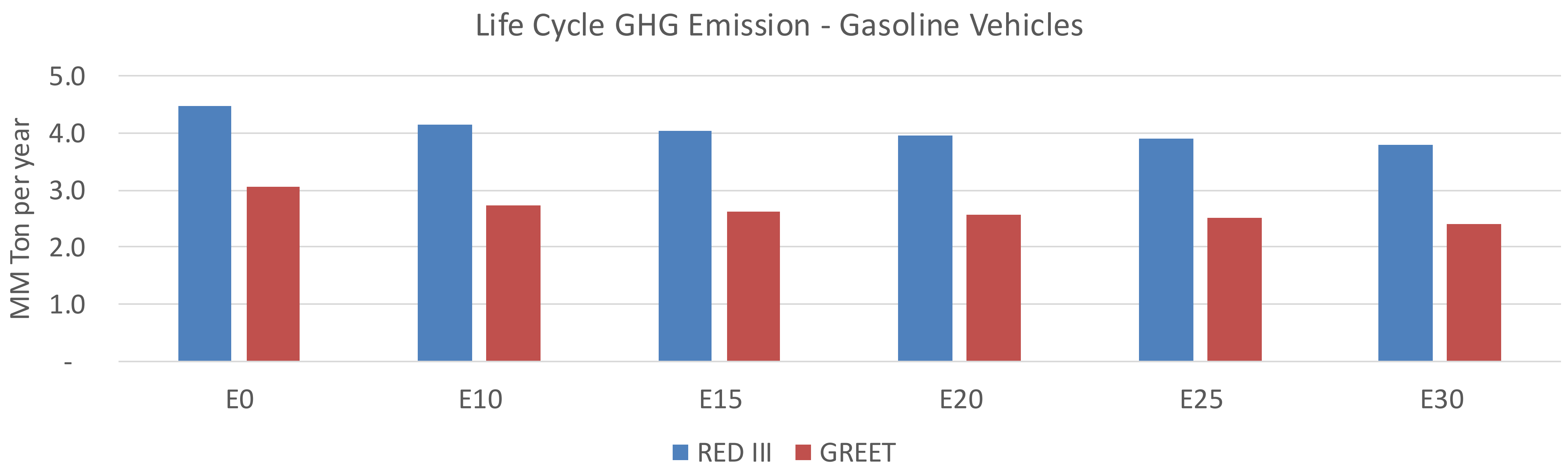
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Slovenia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	16.5
Target @ 2035 (MMT/year)	9.3
Delta	7.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.8%	13.7%	16.0%	18.0%	20.9%
RED II/III	0.0%	7.7%	9.9%	11.6%	13.1%	15.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.6%	5.8%	6.7%	7.6%	8.9%
RED II/III	0.0%	4.8%	6.2%	7.2%	8.1%	9.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90